



Basic Principles of Medicine I
Module: Cardiovascular, Pulmonary, Renal (CPR)
| LECTURE: CPR 05

Imaging of the Heart and Mediastinum

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Objectives

**SOM.MK.I
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CPR.2.AN
AT.1351**

Identify the major anatomical features of the mediastinum, heart and great vessels in radiographic images, CT and MRI.



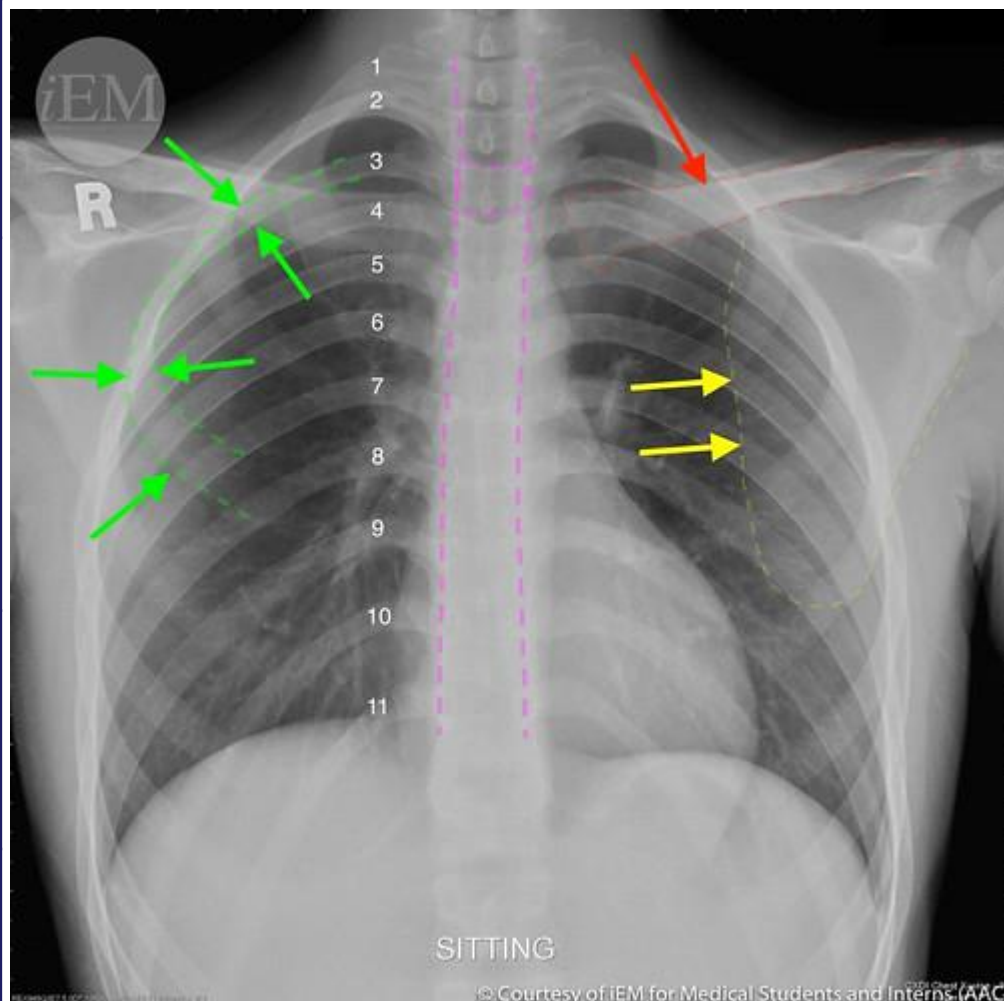
Radiographic Technique

The chest radiograph accounts for approximately 45% of all radiographic examinations in the United States, making it an essential tool in the management of patients with cardiorespiratory disorders.

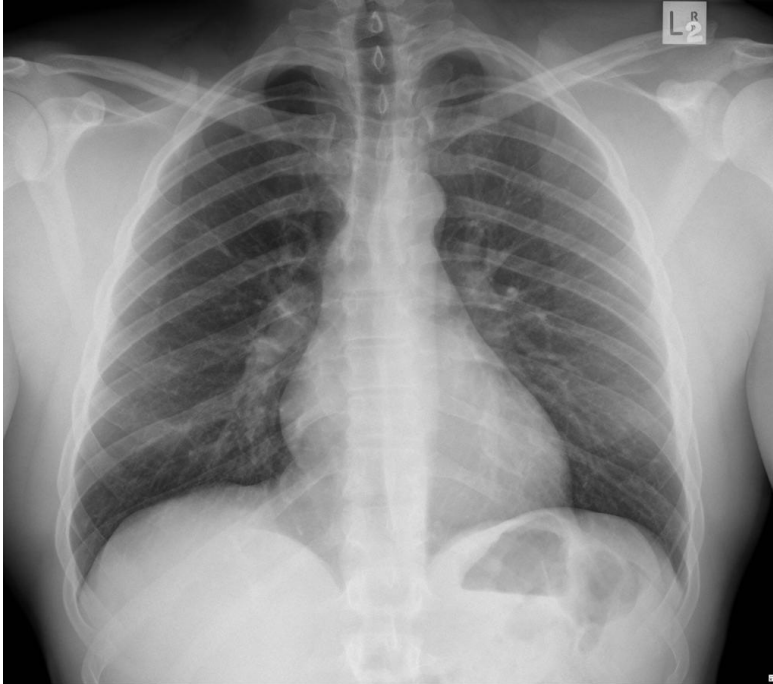
- **The standard chest radiograph** exam consists of two projections, namely **posteroanterior (PA)** and **lateral** views.
- When the patient's clinical condition prevents obtaining a PA view, a single portable **anteroposterior (AP)** view can be done with the understanding that portable films of the chest are less sensitive for detecting disease. This is because of technical limitations such as magnification, suboptimal patient positioning, and variation in exam technique

Technical Adequacy of a PA Chest Film

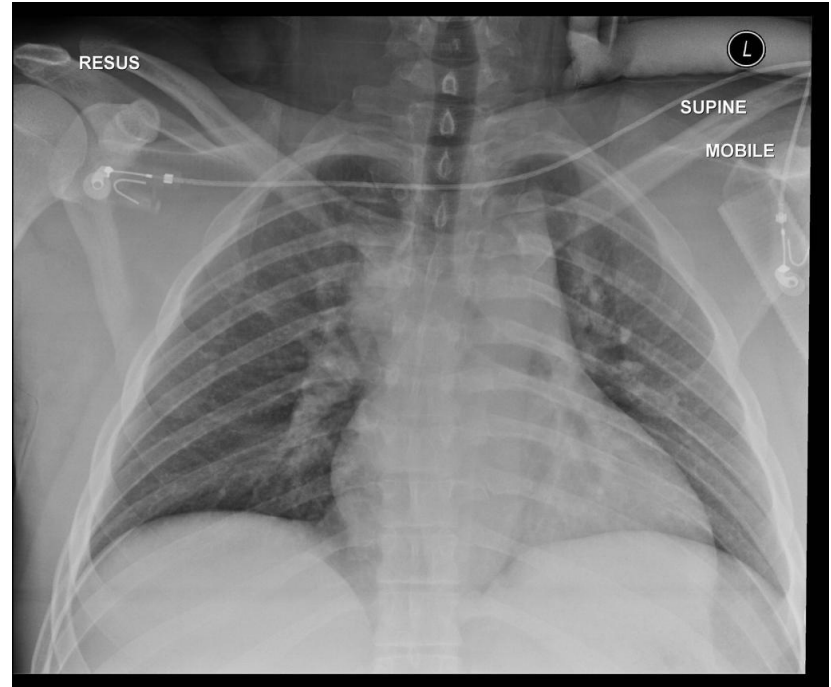
Exposure/penetration	Assessed by looking at the lower thoracic vertebral bodies, whose outline should be barely visible on a PA projection
Inspiratory effort	Anterior aspect of at least six ribs must be noted above the dome of the right diaphragm. Alternatively, posterior aspects of at least eight to nine ribs should be visualized
Rotation	Thoracic spinous processes are equidistant from the medial end of each clavicle on a frontal image
Angulation	Medial ends of the clavicles should be projected over the posterior third or fourth rib



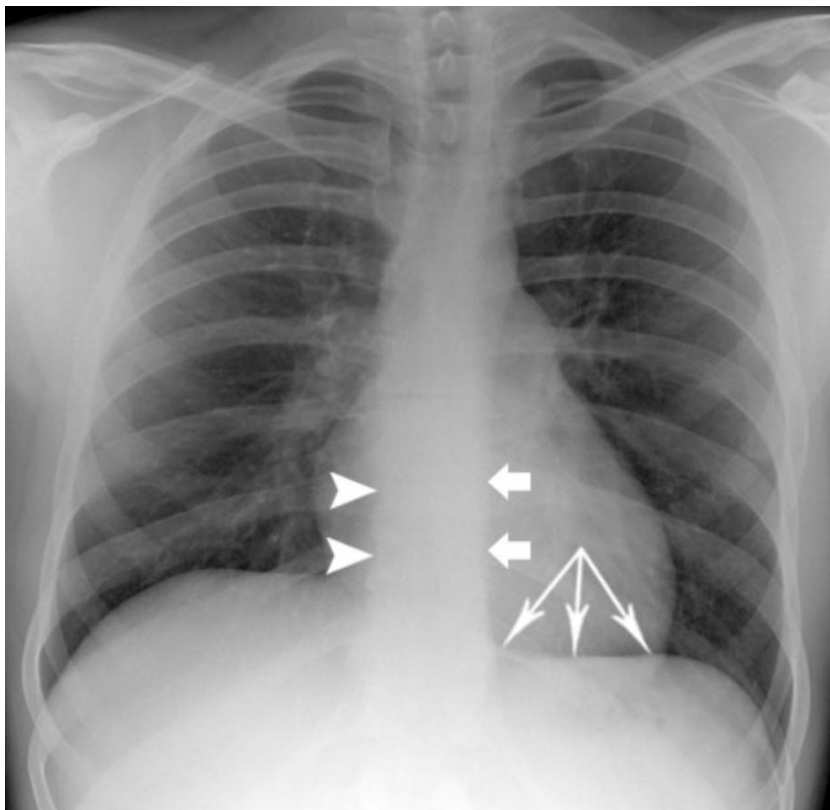
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The standard chest radiograph is acquired with the patient standing up, and with the X-ray beam passing through the patient from Posterior to Anterior (PA).

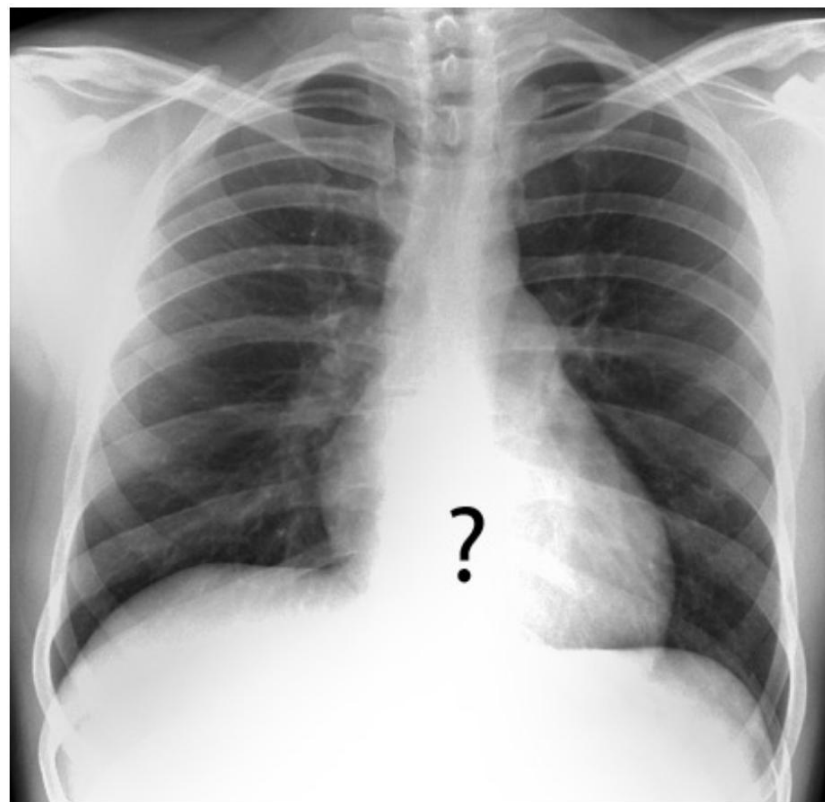


The superior mediastinum appears widened due to AP magnification.
The heart appears enlarged - a combination of AP magnification and underinflation
There appears to be a bilateral interstitial infiltrate/airways thickening - probably also due to underinflation.



Good Quality

- The diaphragm (**long arrows**) is visible to the spine.
- The left paravertebral soft tissues are visible (**short arrows**), and the right side of the spine is clear (**arrowheads**).
- There is no abnormality in lung tissue behind the heart.



Underpenetration

- The left hemidiaphragm is not visible to the spine
- Lung tissue behind the heart cannot be assessed
- Re-windowing the image using digital software can compensate



ABCDE approach

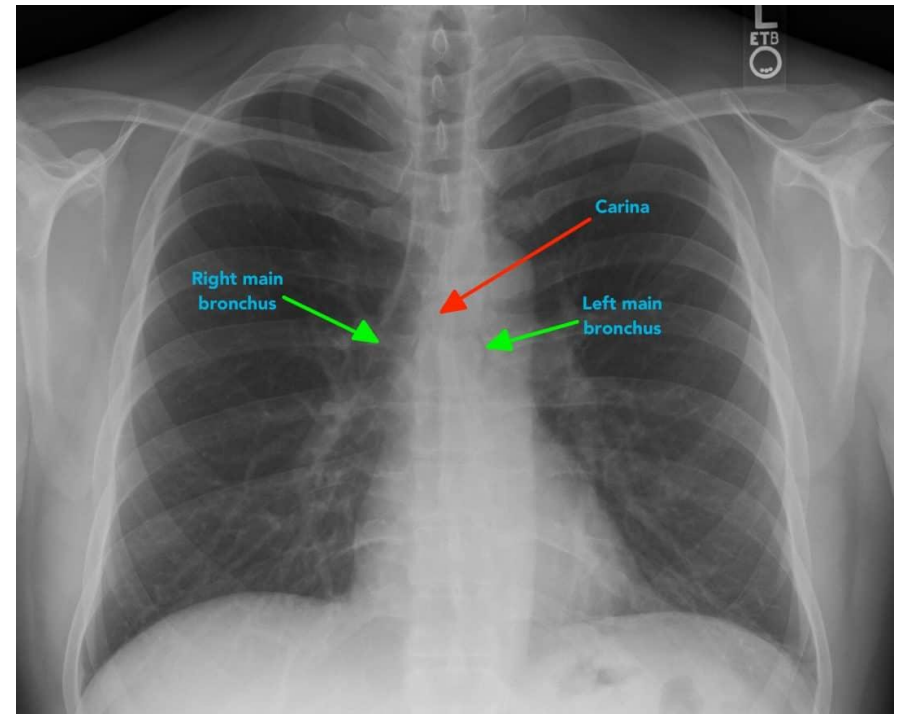
- **A**irway: trachea, carina, bronchi and hilar structures.
- **B**reathing: lungs and pleura.
- **C**ardiac: heart size and borders.
- **D**iaphragm: including assessment of costophrenic angles.
- **E**verything else: mediastinal contours, bones, soft tissues, tubes, valves, pacemakers and review areas.

Airway - Trachea

- Inspect the trachea for evidence of **deviation**:
- The trachea is normally located centrally or deviating very slightly to the right.
 - **True tracheal deviation**:
 - Pushing of the trachea: large pleural effusion or tension pneumothorax.
 - Pulling of the trachea: consolidation with associated lobar collapse.

Airway - Carina and bronchi

- The **carina** is cartilage situated at the point at which the **trachea** divides into the **left** and **right main bronchus**.
- The **right main bronchus** is generally **wider, shorter** and **more vertical** than the **left main bronchus**. As a result of this difference in size and orientation, it is more common for **inhaled foreign objects** to become lodged in the **right main bronchus**.





Airway - Hilar structures

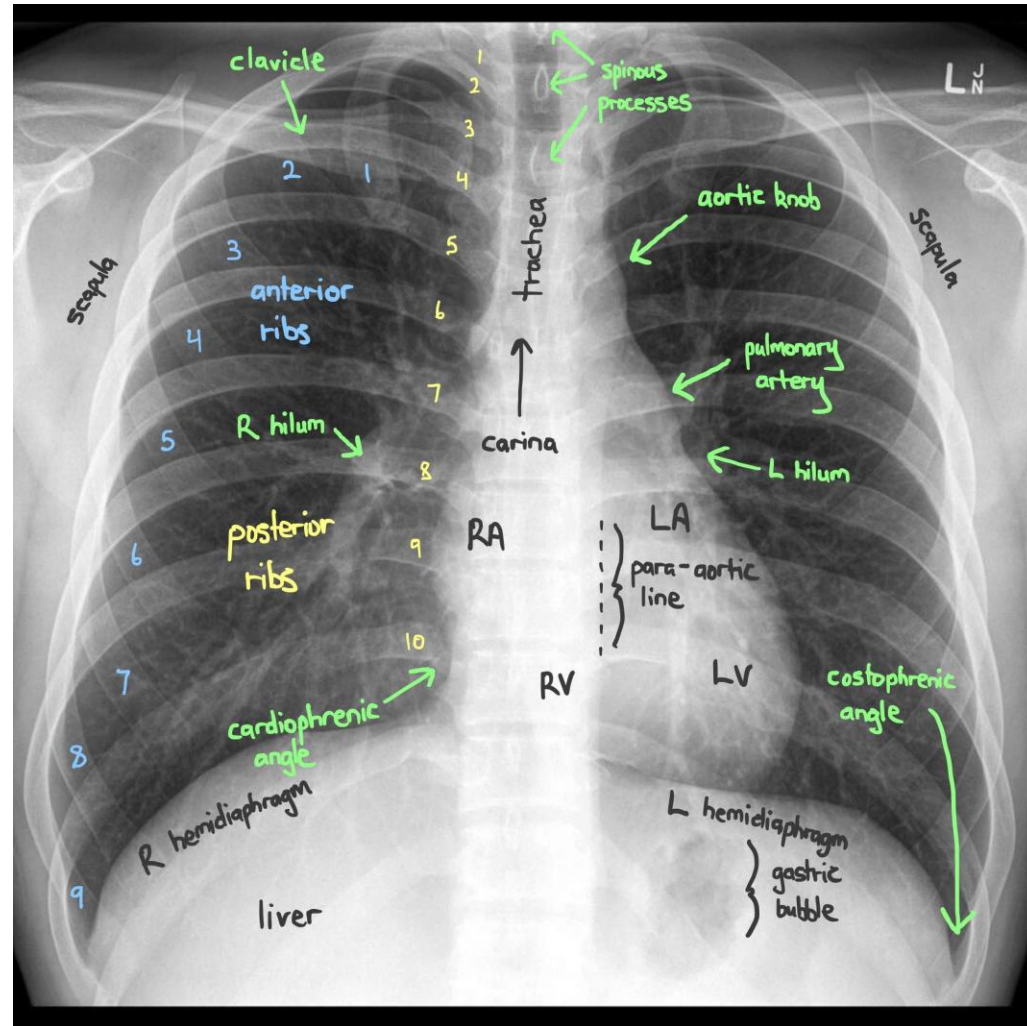
- The **hilar** consist of the **main pulmonary vasculature** and the **major bronchi**.
- Each hilar also has a collection of **lymph nodes** which **are not usually visible** in healthy individuals

Hilar enlargement can be caused by several different pathologies:

- Bilateral symmetrical enlargement is typically associated with sarcoidosis.
- Unilateral/asymmetrical enlargement may be due to underlying malignancy.

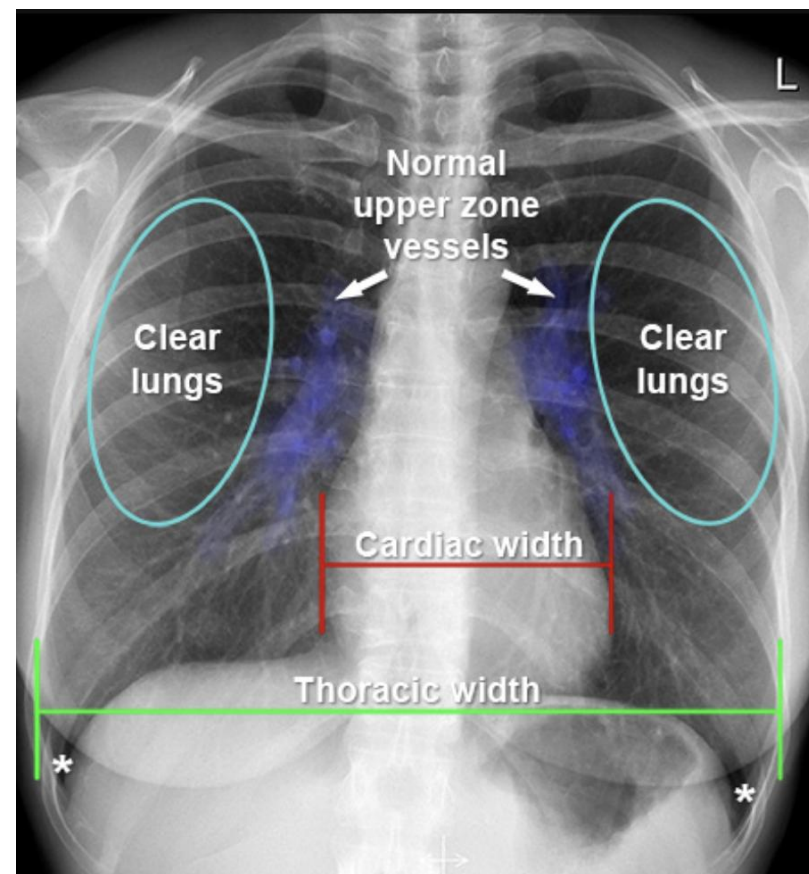
Breathing

- Both lungs should be well-expanded and similar in volume
 - can you count 10 posterior ribs bilaterally?
- Compare the apical, upper, middle, and lower zones in turn
 - are there areas of increased density?
- Trace the lung vessels
 - do they branch out progressively and uniformly?
 - are there extra lines in the periphery that are not vessels?
- Trace the lateral margins of the lung to the costophrenic angles
- Trace the hemidiaphragms into the vertebra
- Trace the cardiac borders
 - can you clearly see the left and right heart border?
 - can you see the descending aorta?



Cardiac - Assess heart size

- In a healthy individual, the **heart** should **occupy no more than 50% of the thoracic width** (i.e. a cardiothoracic ratio of less than 0.5).
- This rule only applies to **PA chest X-rays** (as AP films exaggerate heart size), so you should not draw any conclusions about heart size from an AP film.



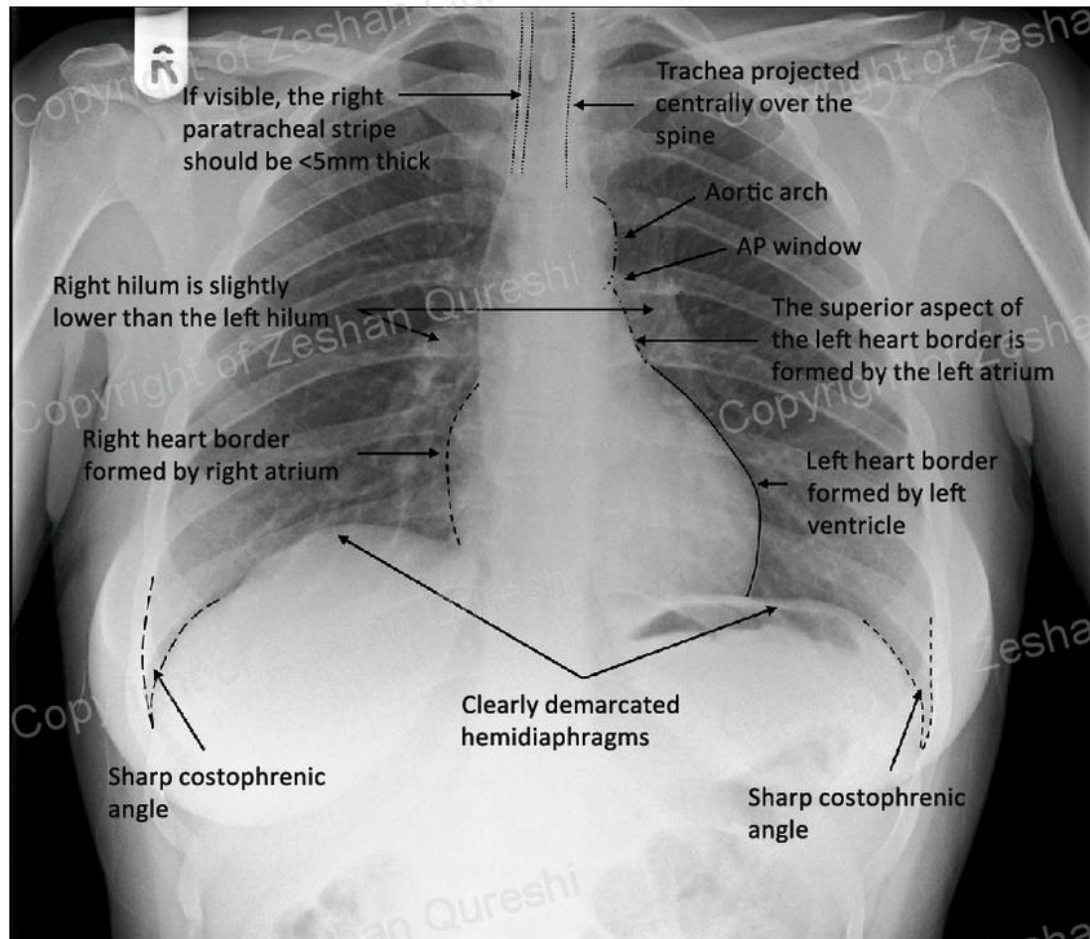
In most normal adults at **full inspiration**, the cardiothoracic ratio is **less than 50%**. That is, the size of the heart is usually less than half of the internal diameter of the thoracic rib cage on a standard PA frontal radiograph taken with an adequate inspiration (about nine posterior ribs showing).



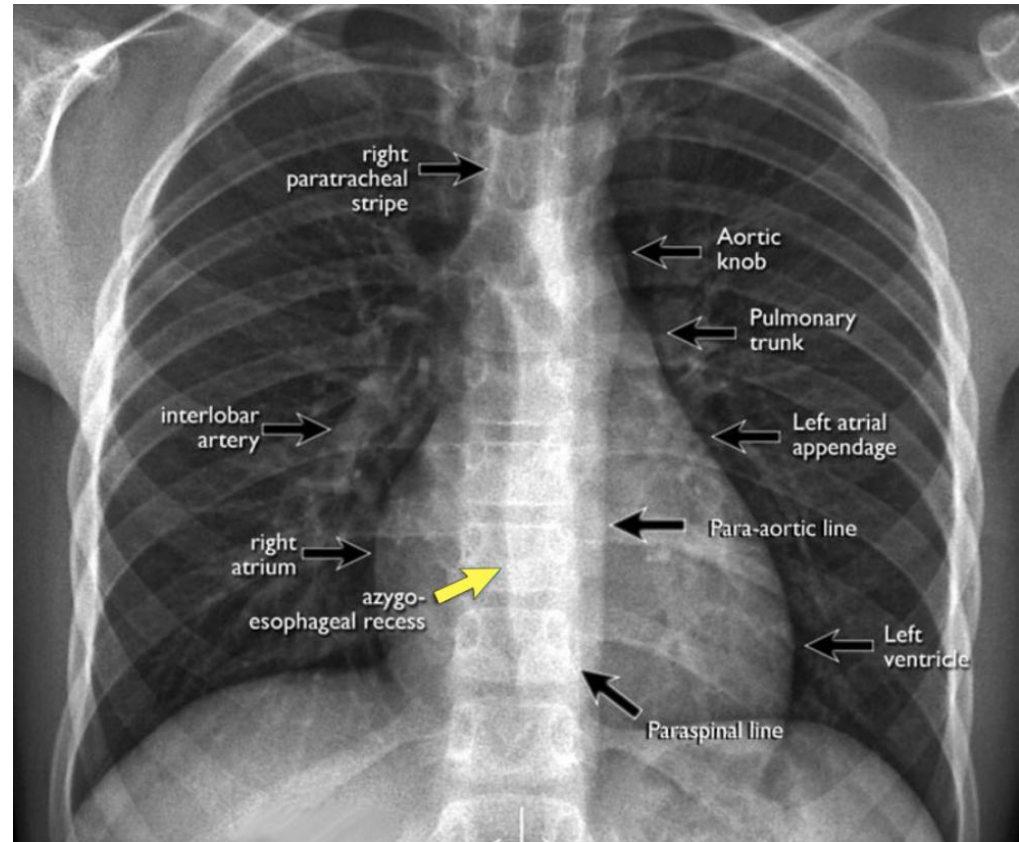
Cardiac - Assess the heart's borders

- Inspect the **borders of the heart** which should be well defined in healthy individuals:
 - The right atrium makes up most of the right heart border.
 - The left ventricle makes up most of the left heart border.

- On the right side of the heart, the first contour is the **ascending aorta**.
- The right heart border is formed by the **right atrium**.
- On the left, the first contour is the **aortic knob**, a radiographic structure formed by the foreshortened aortic arch superimposed on a portion of the proximal descending aorta.



- The next contour below the aortic knob is the **main pulmonary artery** before it divides into a right and left pulmonary artery.
- Just below the main pulmonary artery segment, there is normally a **slight indentation** where an enlarged left atrium/left atrial appendage may appear on the left side of the heart.
- The last contour of the heart on the left is formed by the **left ventricle**. The descending aorta almost disappears with the shadow of the spine.



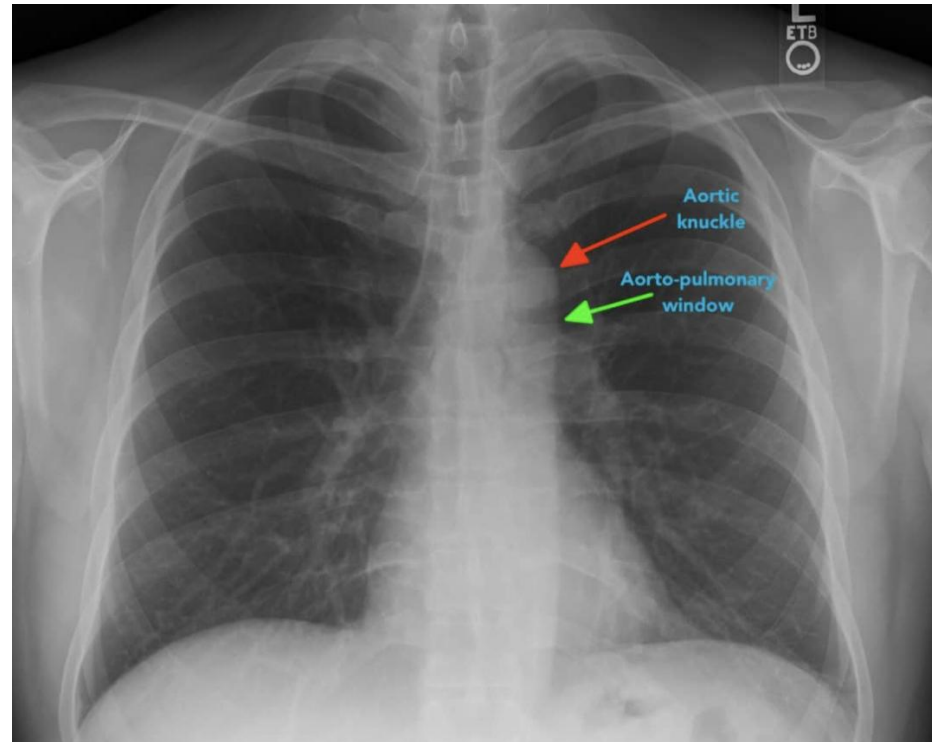


Diaphragm

- The **right hemidiaphragm** is, in most cases, **higher than the left** in healthy individuals (due to the presence of the liver). The stomach underlies the left hemidiaphragm and is best identified by the gastric bubble located within it.

Everything else - Mediastinal contours

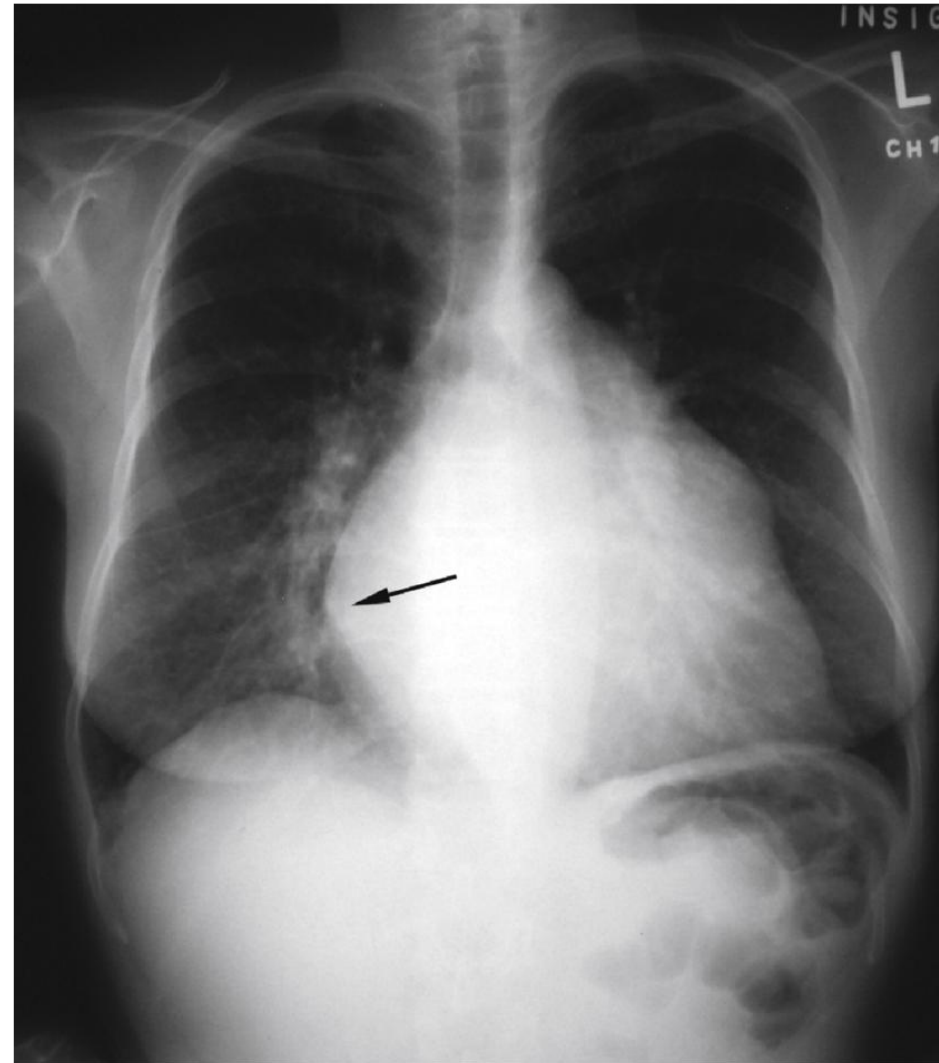
- The **mediastinum** contains the **heart**, **great vessels**, **lymphoid tissue** and a number of **potential spaces** where pathology can develop.
- **Aortic knuckle**
 - The **aortic knuckle** is located at the **left lateral edge of the aorta** as it arches back over the left main bronchus. Reduced definition of the aortic knuckle contours can occur in the context of an **aneurysm**.
- **Aortopulmonary window**
 - The **aortopulmonary window** is a space located **between the arch of the aorta and the pulmonary arteries**. This space can be lost as a result of **mediastinal lymphadenopathy** (e.g. malignancy).



Case study

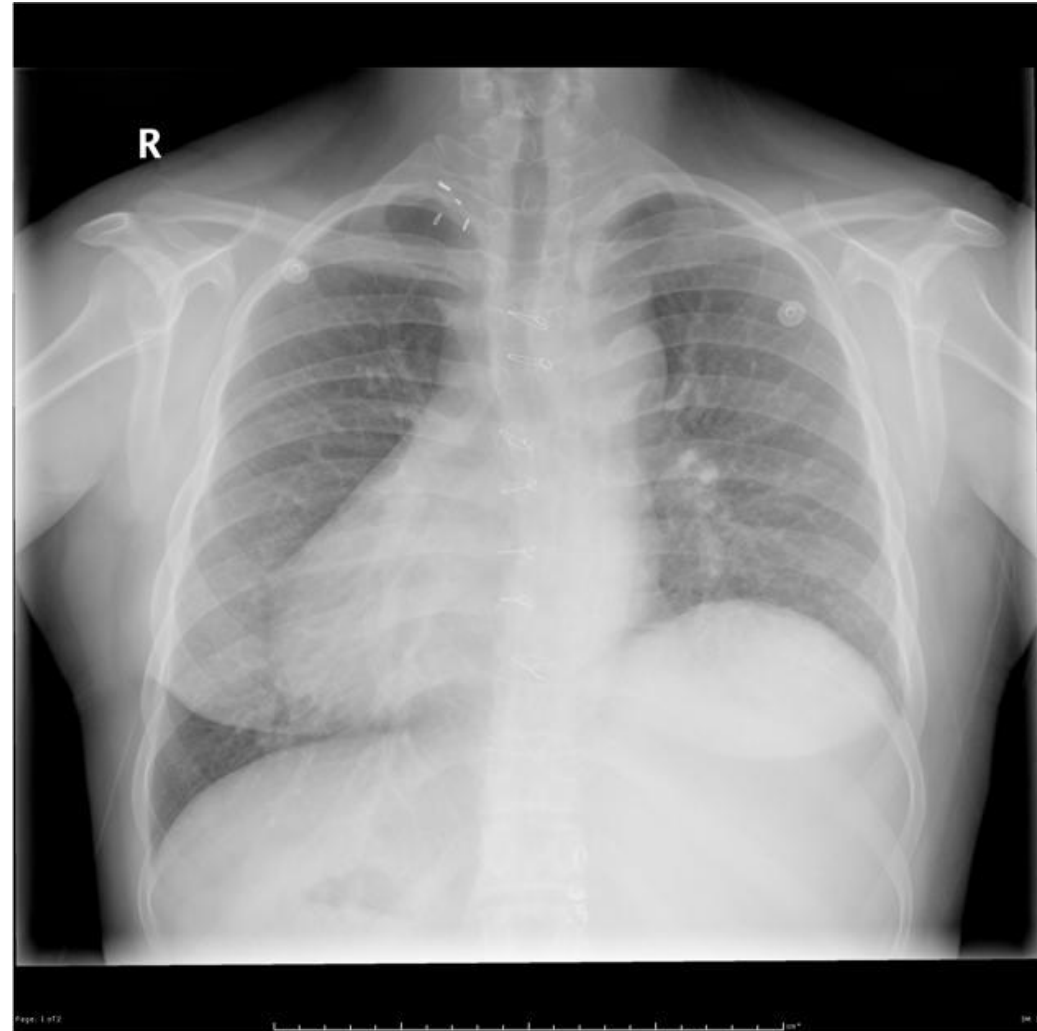
A 23-year-old man came to the physician because of shortness of breath and lower extremity edema for the last few weeks. Physical examination shows apical pan-systolic murmur on auscultation.

This film shows the double density sign (*arrow*) produced by the overlapping of the left atrium with the right heart border (right atrium). Also, note the outward convexity of the upper left heart border typically seen with left atrial enlargement.



Case study

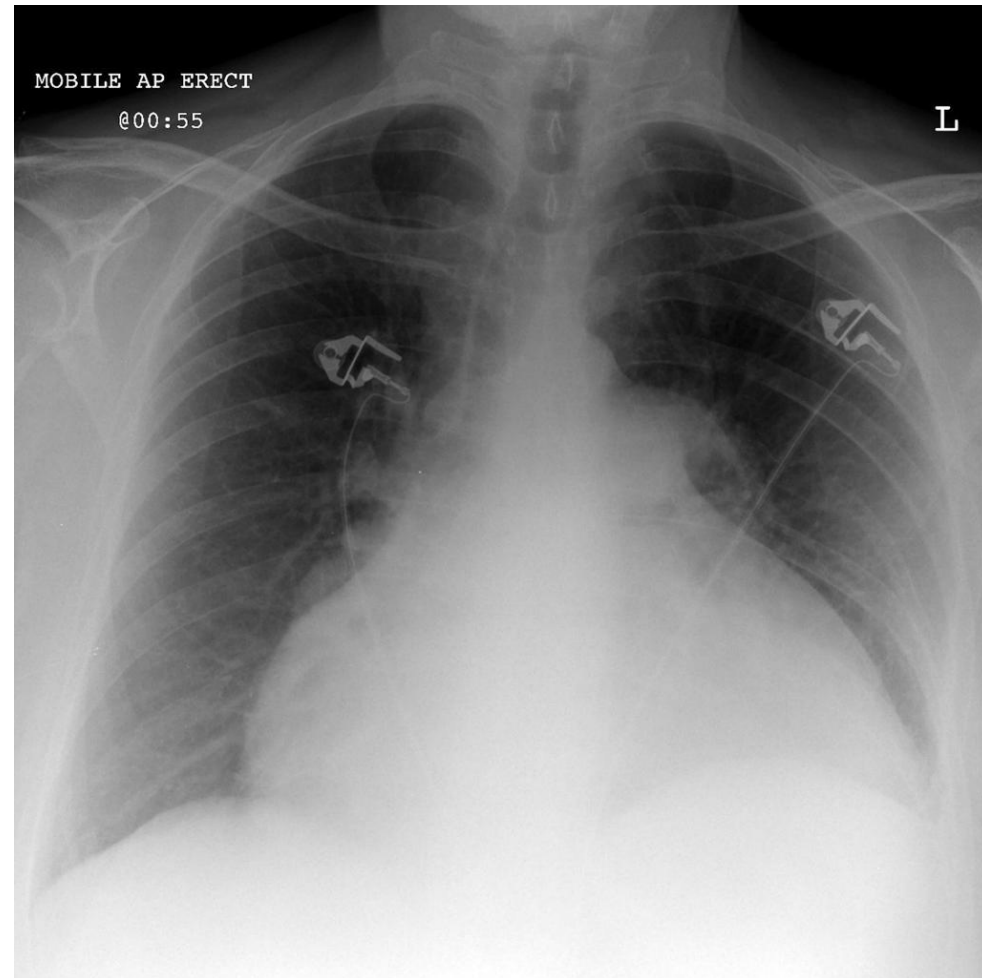
A 24-year-old man came to the physician because of constant chest pain for the last 2 days. He said the pain doesn't increase or decrease with respiration or exercise. Physical examinations show the absence of the apical impulse and heart sounds on the right side of the chest. A chest X-ray was requested.



Case study

A 26-year-old woman presented to the emergency department because of sharp left chest pain which radiate to the lower end of the scapula for the last few hours. He said the pain gets worse when he is in a supine position. She has a 3-day history of upper respiratory symptoms.

Physical examination shows pulsus paradoxus, muffled heart sounds, and friction rub on the left border of the sternum.



Pericardial effusion - water bottle sign

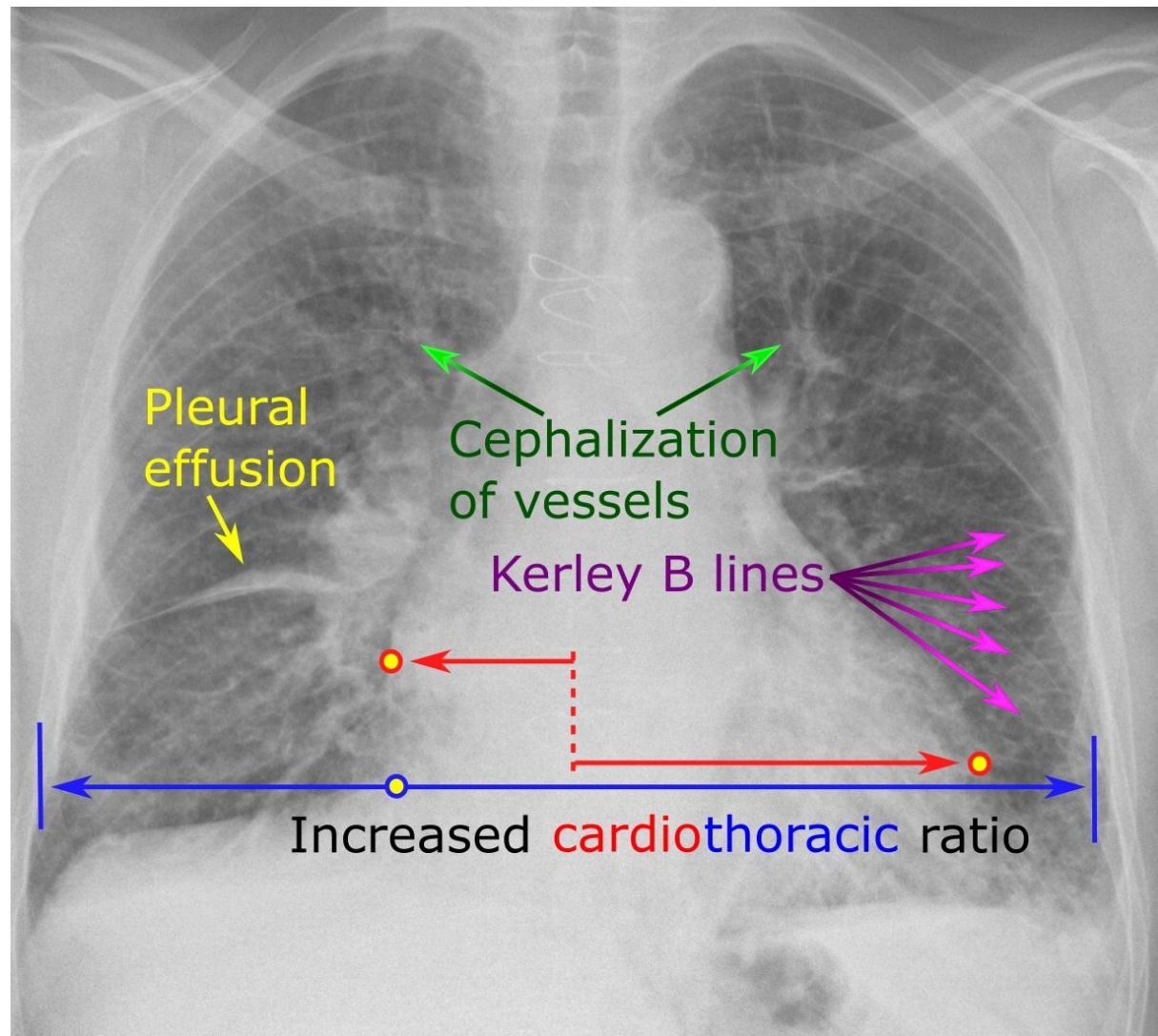
Case study

- A 54-year-old woman presented to the emergency department with shortness of breath (SOB) for 4 months that progressed to an extent that she was unable to perform daily activities. She also used 3 pillows to sleep and often woke up from sleep due to difficulty catching her breath. Her medical history included hypertension, dyslipidemia, diabetes mellitus, and a history of triple bypass surgery 4 years ago. Her current home medications included aspirin, atorvastatin, amlodipine, and metformin. Physical examination showed bilateral diffuse crackles in the lungs, elevated jugular venous pressure, and 2+ pitting lower extremity edema. ECG showed normal sinus rhythm with left ventricular hypertrophy.



A good mnemonic to remember these principles is ABCDE:

- A - alveolar edema (bat wing opacities)
- B - Kerley B lines
- C - cardiomegaly
- D - dilated upper lobe vessels (Cephalization of vessels)
- E - pleural effusion

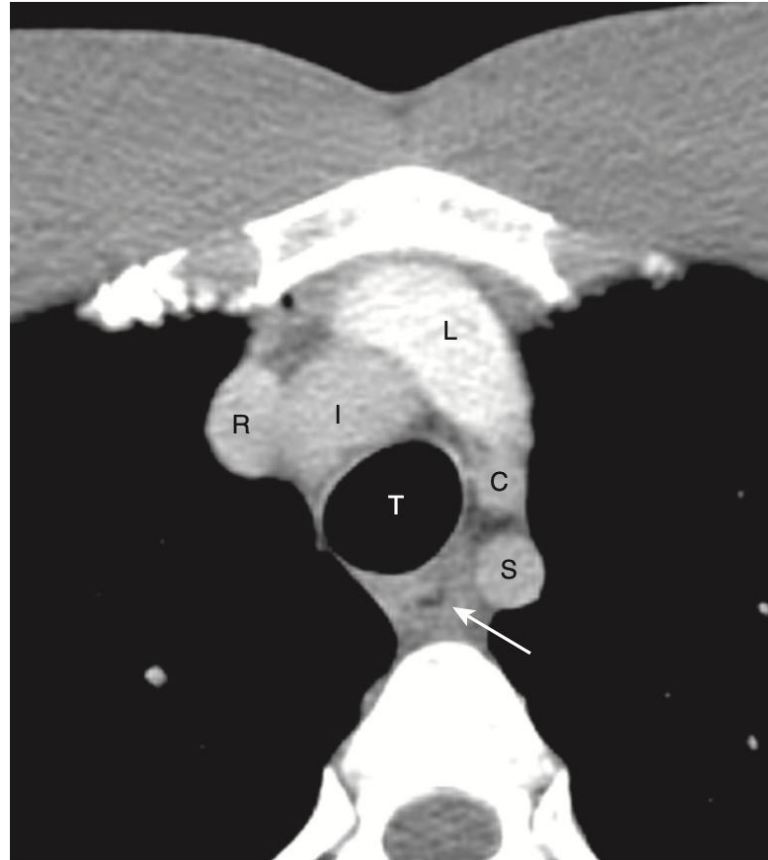




Normal Cardiac CT Anatomy

- We will start at the top of the chest and progress inferiorly, highlighting the major structures visible at six key levels.
 1. Five-Vessel Level
 2. Aortic Arch Level
 3. Aortopulmonary Window Level
 4. Main Pulmonary Artery Level
 5. High Cardiac Level
 6. Low Cardiac Level

Five-Vessel Level



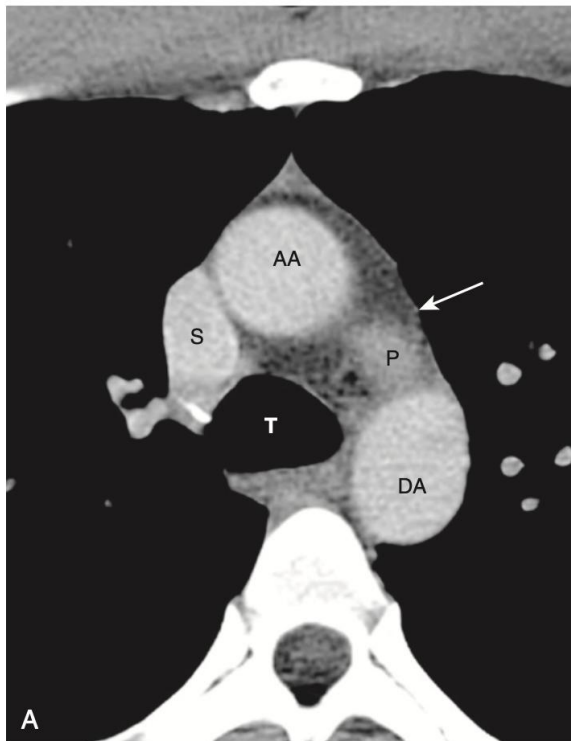
At this level, you should be able to identify the lungs, the **trachea (T)** and the **esophagus** (*white arrow*). The **right brachiocephalic vein (R)** is the vessel to the right of the **trachea (T)**. The **left brachiocephalic vein (L)** lies just posterior to the sternum. From the patient's right to the patient's left, the arteries you see may include the **innominate (brachiocephalic) artery (I)**, **left common carotid (C)**, and **left subclavian arteries (S)**.

Aortic Arch Level



At this level, you should be able to identify the **aortic arch (AA)**, **superior vena cava (S)**, and **azygos vein**

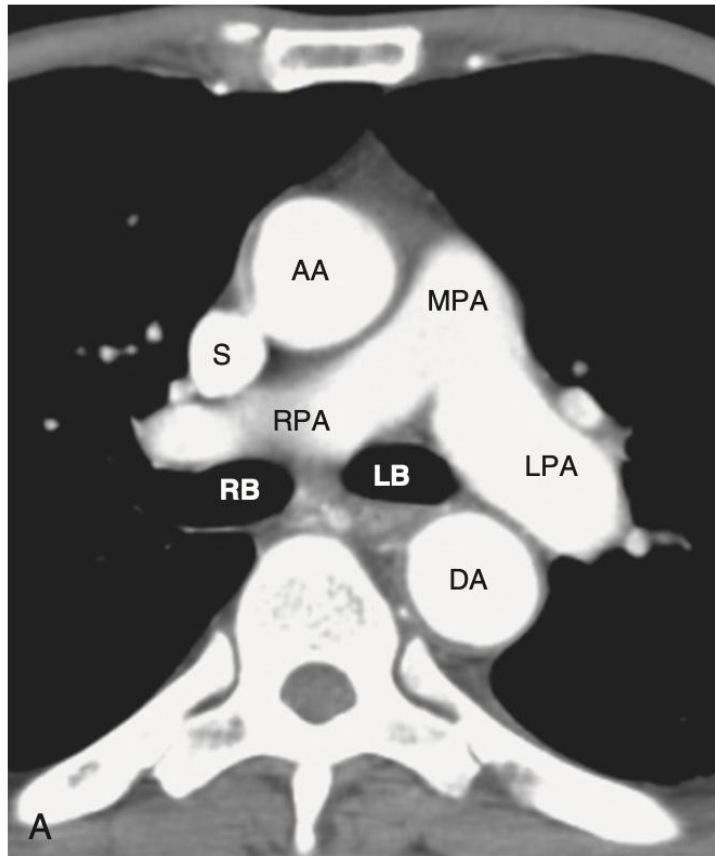
Aortopulmonary Window Level



At this level you should be able to identify the **trachea (T)**, **ascending (AA)** and **descending aorta (DA)**, **superior vena cava (S)**, and possibly the uppermost aspect of the **left pulmonary artery (P)**. In most people, there is a space visible just under the arch of the aorta and above the pulmonary artery called the **aortopulmonary window** (*white arrow*).

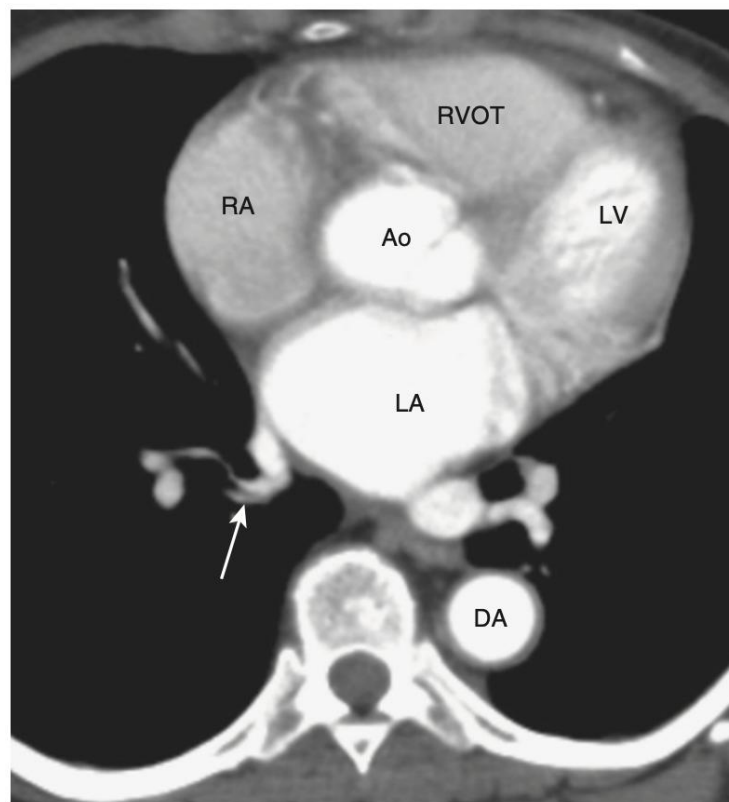
(B) Coronal reformatted CT enables us to also see the **right atrium (RA)**, **superior vena cava (S)**, **pulmonary artery (P)**, **left ventricle (LV)**, **aortic valve** (*black arrow*), **left atrial appendage** (*white arrow*), and origin of the great vessels (*white circle*).

Main Pulmonary Artery Level



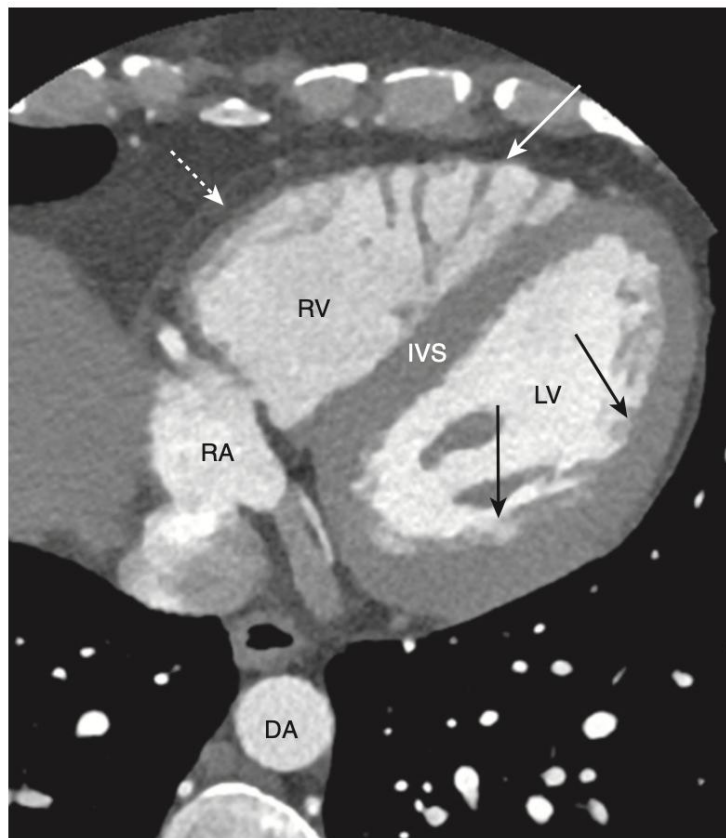
At this level, you should be able to identify the **main (MPA)**, **right (RPA)**, and **left pulmonary arteries (LPA)**, the **right (RB)** and **left main bronchi (LB)**, and the **superior vena cava (S)**. The left pulmonary artery passes anterior to the **descending aorta (DA)**. The right pulmonary artery passes posterior to the **ascending aorta (AA)** and crosses to the right side.

High Cardiac Level



At this level, you should be able to identify the **left atrium (LA)**, **right atrium (RA)**, **aortic root (Ao)**, and **right ventricular outflow tract (RVOT)**. The **left atrium** occupies the posterior and central portion of the heart. One or more **pulmonary veins** may be seen to enter the left atrium (*white arrow*). The right atrium produces the right heart border and lies anteriorly and to the right of the left atrium. DA, Descending thoracic aorta; LV, Left ventricle.

Low Cardiac Level



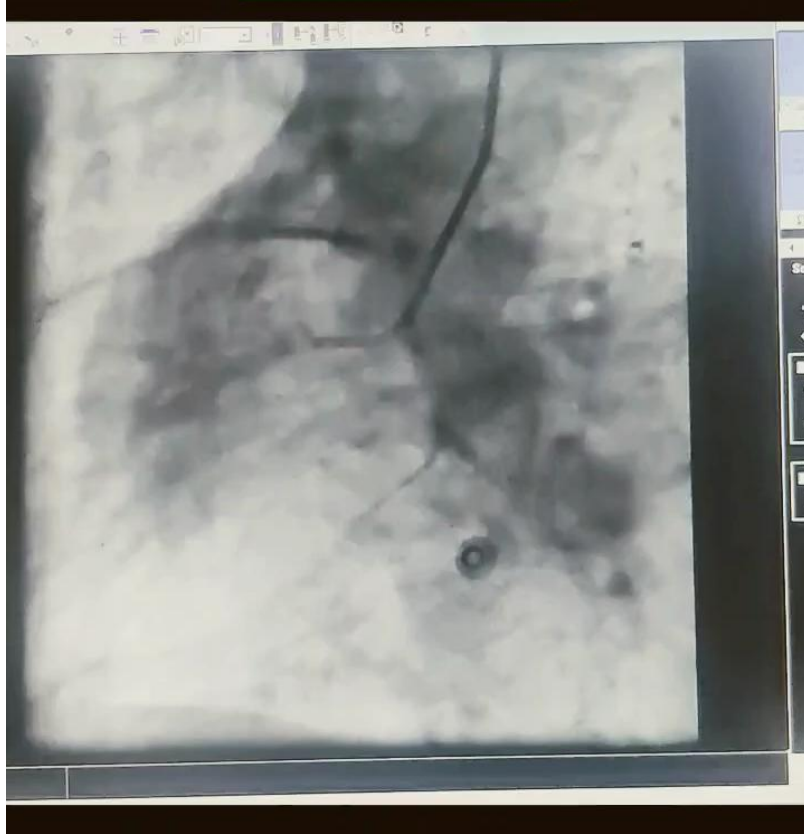
At this level, you should be able to identify the **right atrium (RA)**, **right ventricle (RV)**, **left ventricle (LV)**, and **interventricular septum (IVS)**. The right ventricle is more **heavily trabeculated** (*solid white arrow*) and has a **thinner wall** (*dotted white arrow*) than the wall of the left ventricle (*solid black arrows*). *DA*, Descending thoracic aorta.

Coronary CT Angiography: Normal Anatomy

- Coronary CT angiography (CCTA) compares favorably in accuracy with invasive (catheter) coronary angiography, long held as the reference standard in studying the coronary arteries.
- Normal coronary artery anatomy has many variations.



RCA



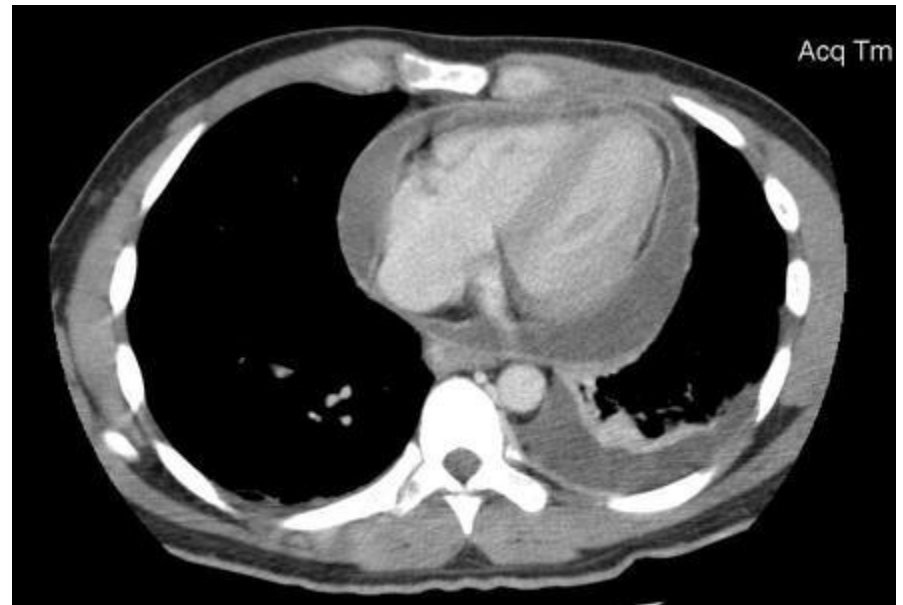
LAD



Pericardial Effusion

Normally, there are 15 to 50 mL of fluid in the pericardial space between the parietal and visceral pericardial layers.

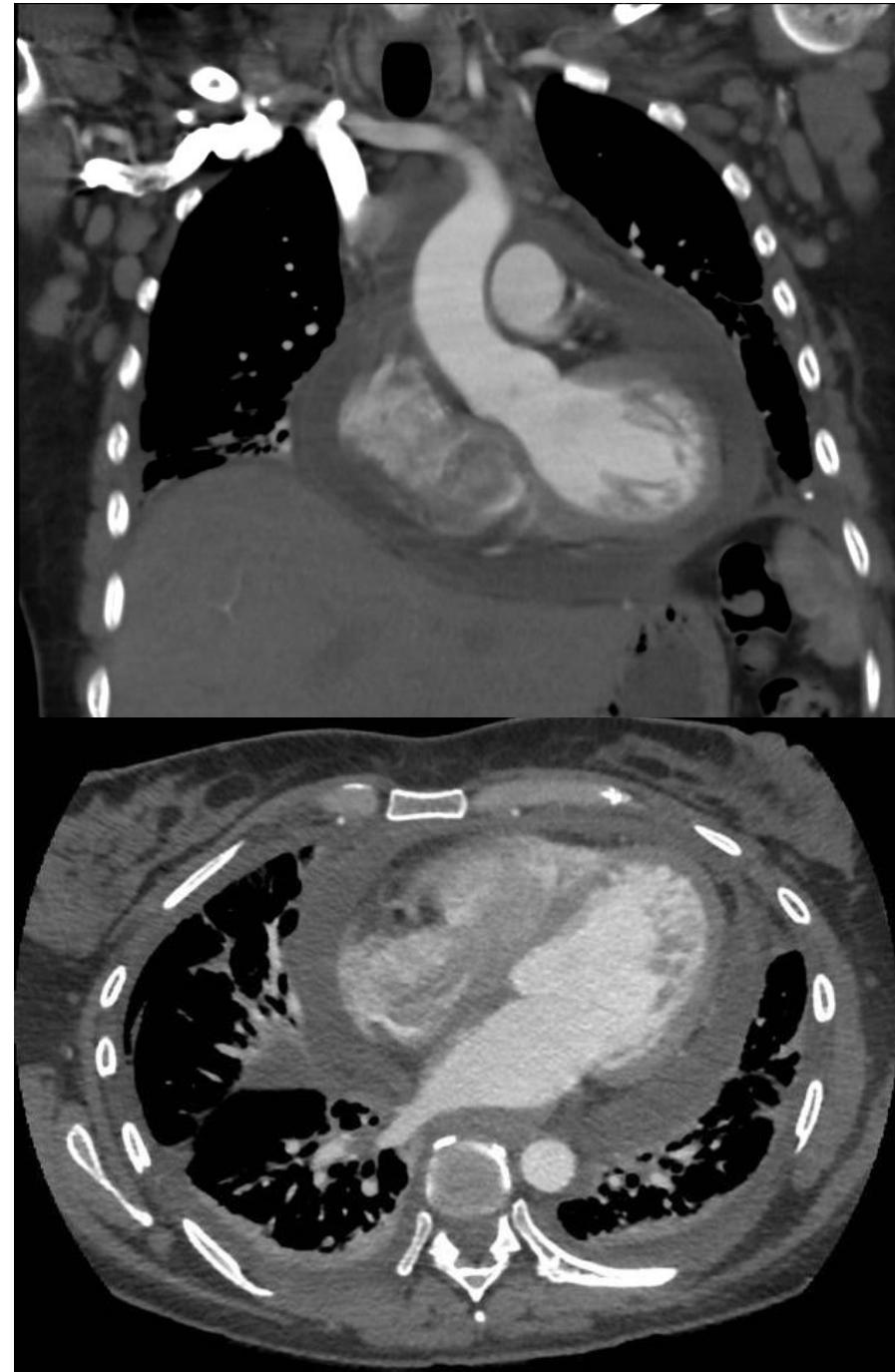
Computed tomography (CT) scans can demonstrate small pericardial effusions, although pericardial ultrasonography is usually the imaging study of first choice. Conventional radiographs are poor at defining a pericardial effusion.



Pericardial Effusion

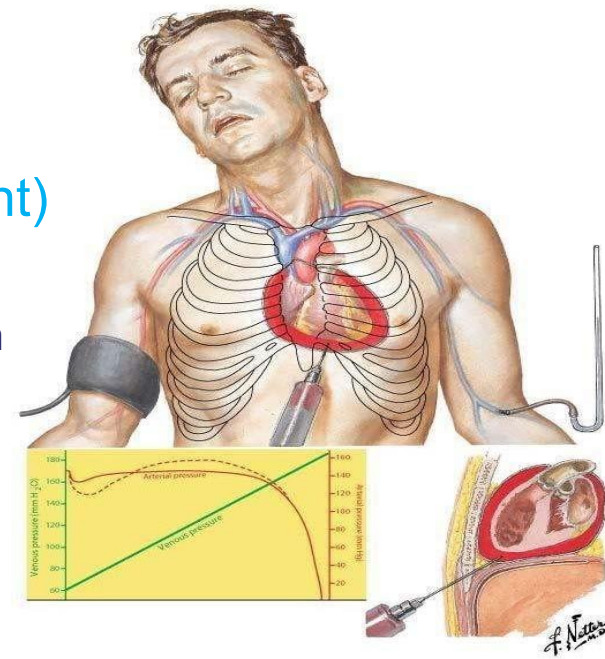
- **May be slow or rapid**
- Many possible causes
 - Systemic inflammatory conditions (Connective tissue disease)
 - Metastasis
 - Hypothyroidism
 - Renal disease
 - Infection
- Symptoms
 - Dyspnea
 - Reduced exercise tolerance
 - Eventually progress to impaired cardiac function
 - May develop into tamponade

[Case courtesy of Dr Mahomed A. Osman, Radiopaedia.org, rID:](#)



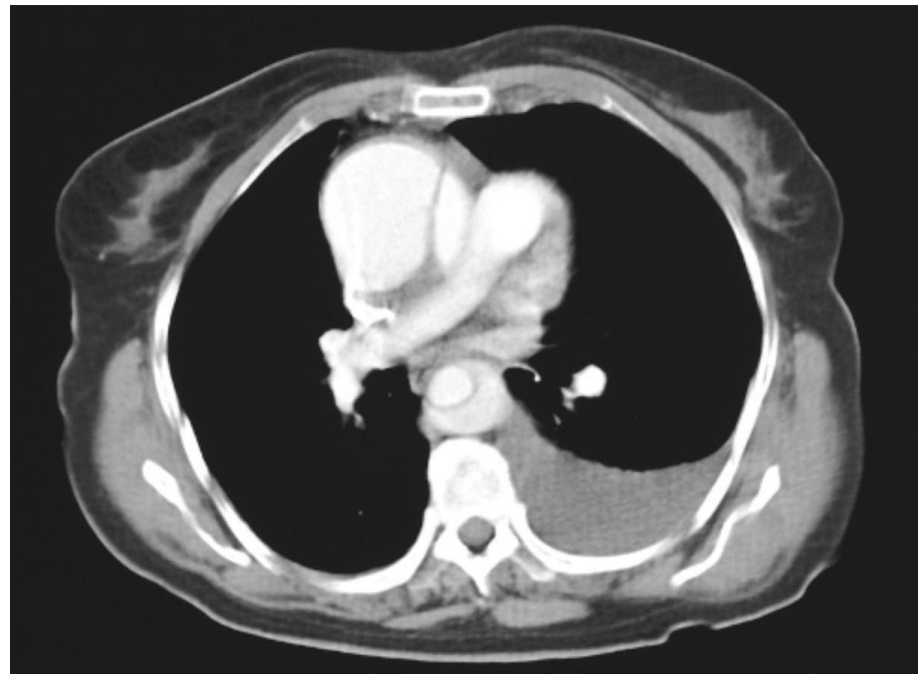
Cardiac Tamponade

- Impaired cardiac output due to an accumulation of fluid, pus, gas, blood or tissue in the pericardial space
- Pulses paradoxus: exaggerated fall in systolic blood pressure (10mmHg) during inspiration
- Increased pressure outside the heart decreases end-diastolic ventricular filling → decreased stroke volume → **lower systolic blood pressure**
- Increased pressure reduces end-systolic atrial filling capacity → rapid increase and higher pressure in atria → **increased pressure in and distension of the jugular veins**
- Increased fluid → **reduced heart sounds (distant)**
- Treatment is by Pericardiocentesis:
 - Paraxiphoid approach – tip of the xiphoid, or between xiphoid and left costal margin, angled towards left shoulder.
 - Apical approach - left 5th or 6th intercostal space



Case study

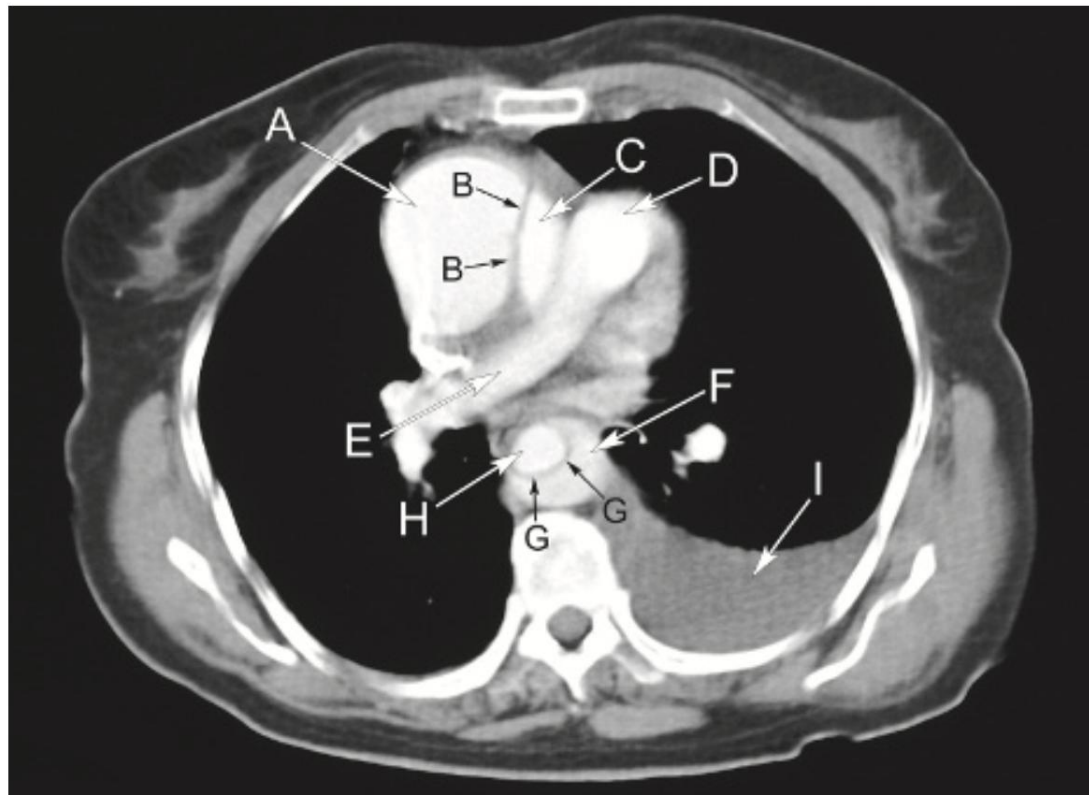
- A 63-year-old male presents to the ED with a severe, constant, sharp, central chest pain radiating to his back, which started 1 hour ago. He is on medication for hypertension and has a 40 pack-year smoking history.
- On examination he looks pale. Heart sounds are normal and his chest is clear. He is tachycardiac at 110 bpm and has a BP in the right arm of 140/96 mmHg and in the left arm of 104/74 mmHg. Radial pulses are present but weaker in the left than the right and femoral pulses also are weak.
- The ECG shows a sinus tachycardia and the CXR reports a small left pleural effusion only. After discussion with a senior colleague, you order a CT thorax



What does the CT thorax show?

What is the likely diagnosis?

This CT demonstrates an aortic dissection with an intimal dissection flap involving both the descending and ascending aorta



Postcontrast CT thorax at the level of the pulmonary trunk
 The ascending aorta true lumen is seen (A) with visible intimal flap (B) and false lumen (C).
 Descending aorta true lumen (F), intimal flap (G) and false lumen (H). Note: left basal pleural effusion (I), normal main pulmonary artery (D), and right main pulmonary artery (E).